

Theorem 5.5 (Multifold sums of powers at zero). *For non-negative integers r, n, m*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \Sigma^s n^0 \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (3.7), and Equation (1). □